

Lab Final Submission - part 1 - Splunk SIEM

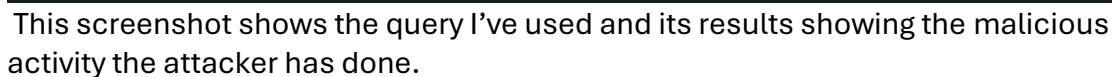
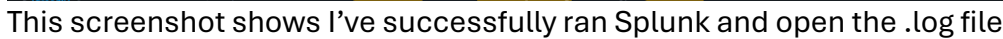
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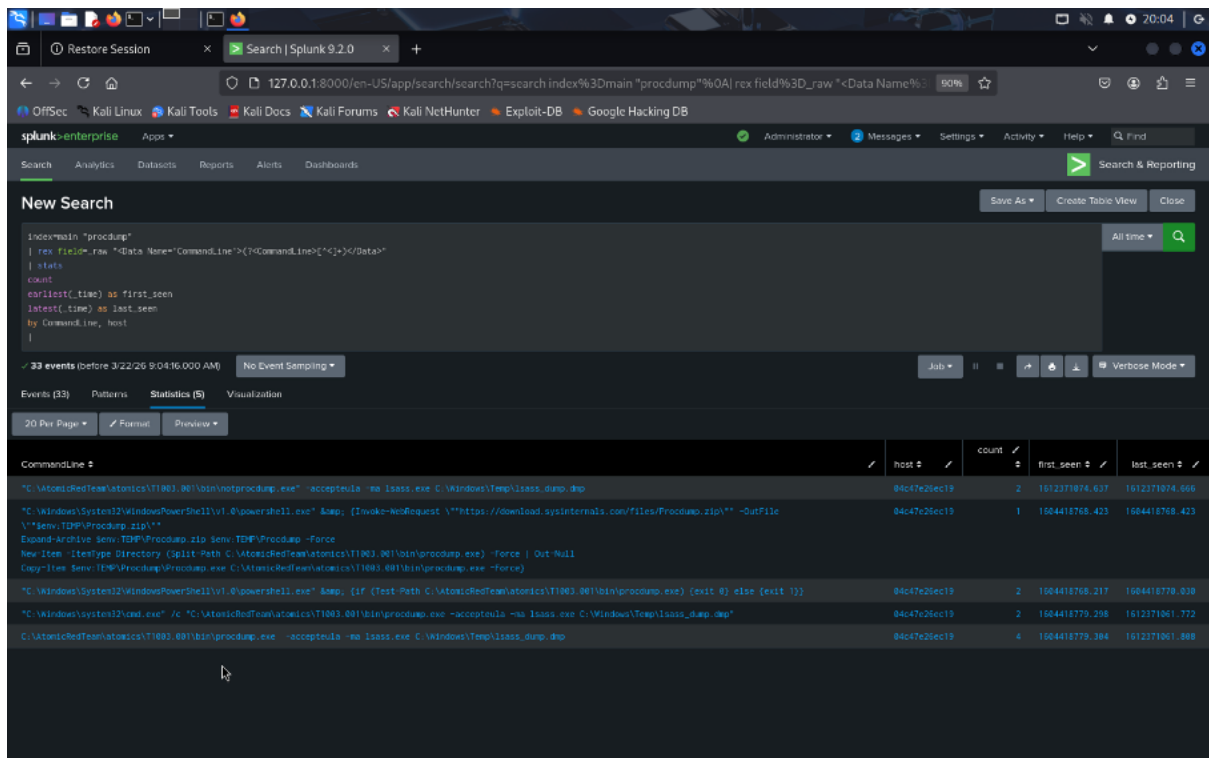
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Section 3

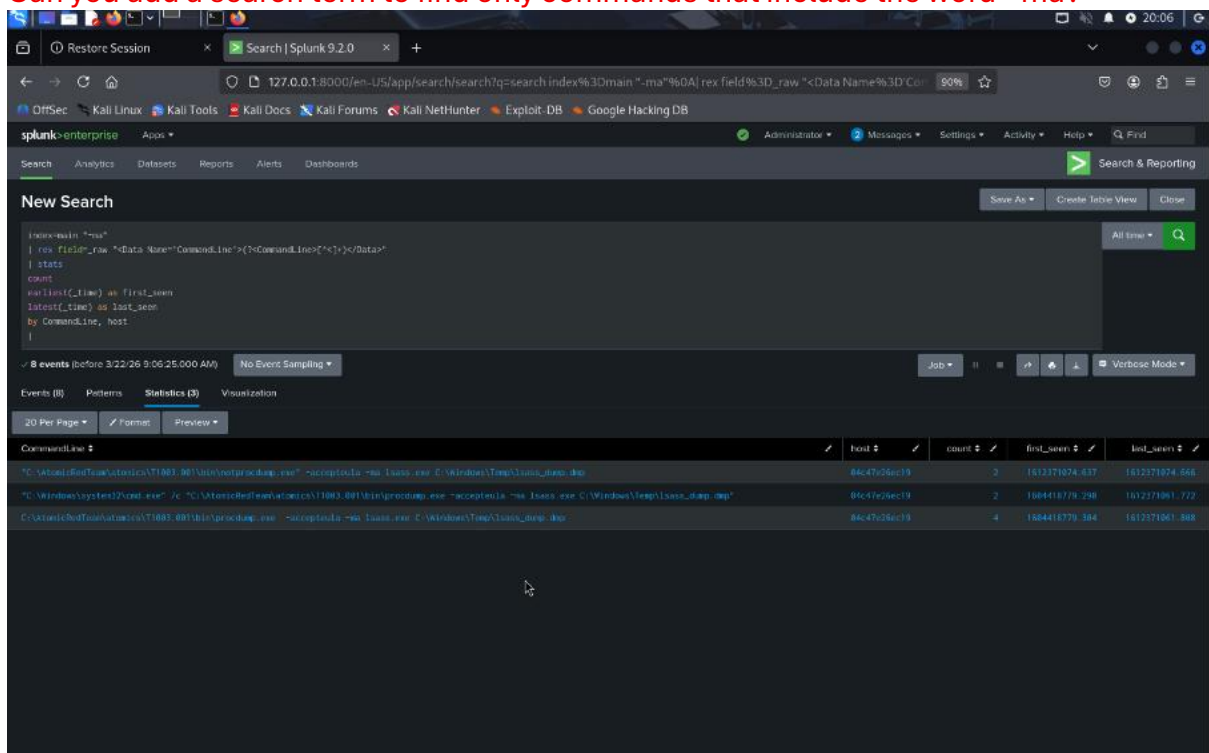


Try removing the | eval lines for example. What happens to the time format?



After removing eval lines the time appears into seconds as we can see in the screenshot.

Can you add a search term to find only commands that include the word - ma?



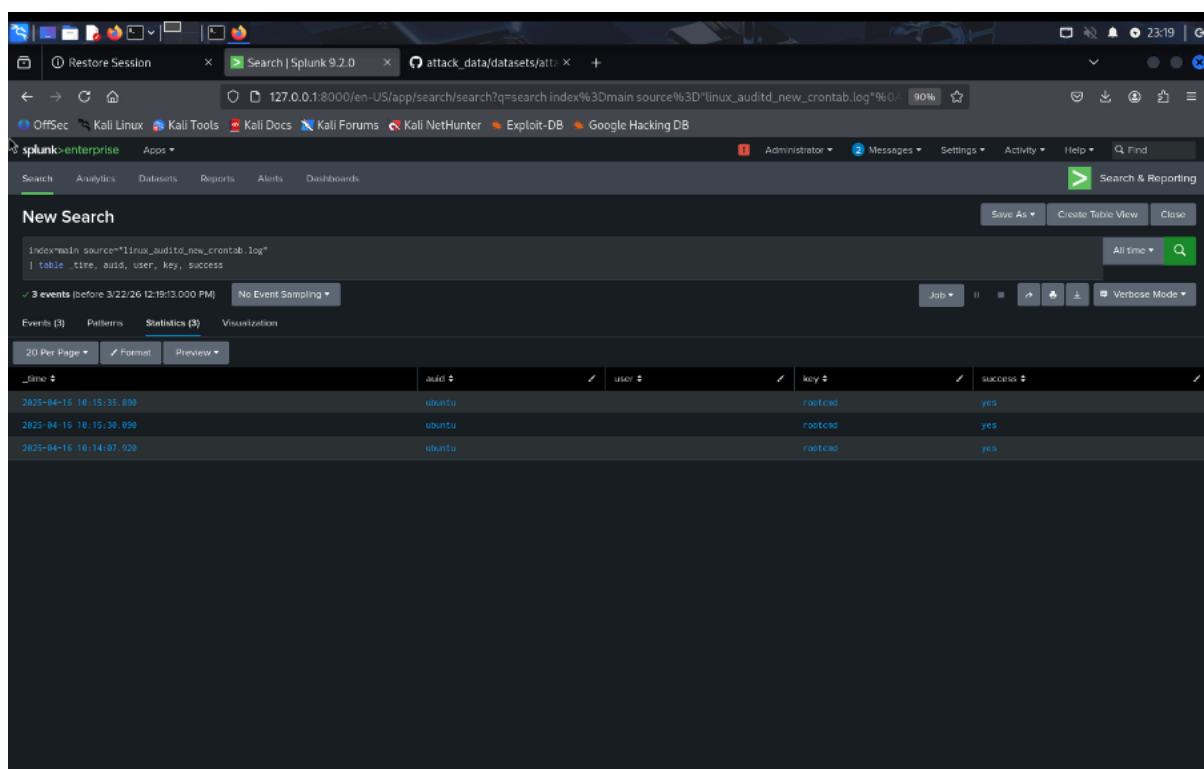
As we can see in the screenshot I've adjust the query to make it shows the commands that has the word 'ma' in it.

Section 4: Reflection and Fostering Independence

Technique Explanation

This attack uses the Linux cron service to schedule malicious tasks. By adding a command to the root crontab, an attacker ensures their script runs automatically with the highest permissions, allowing them to stay in the system even after a reboot.

Screenshot A

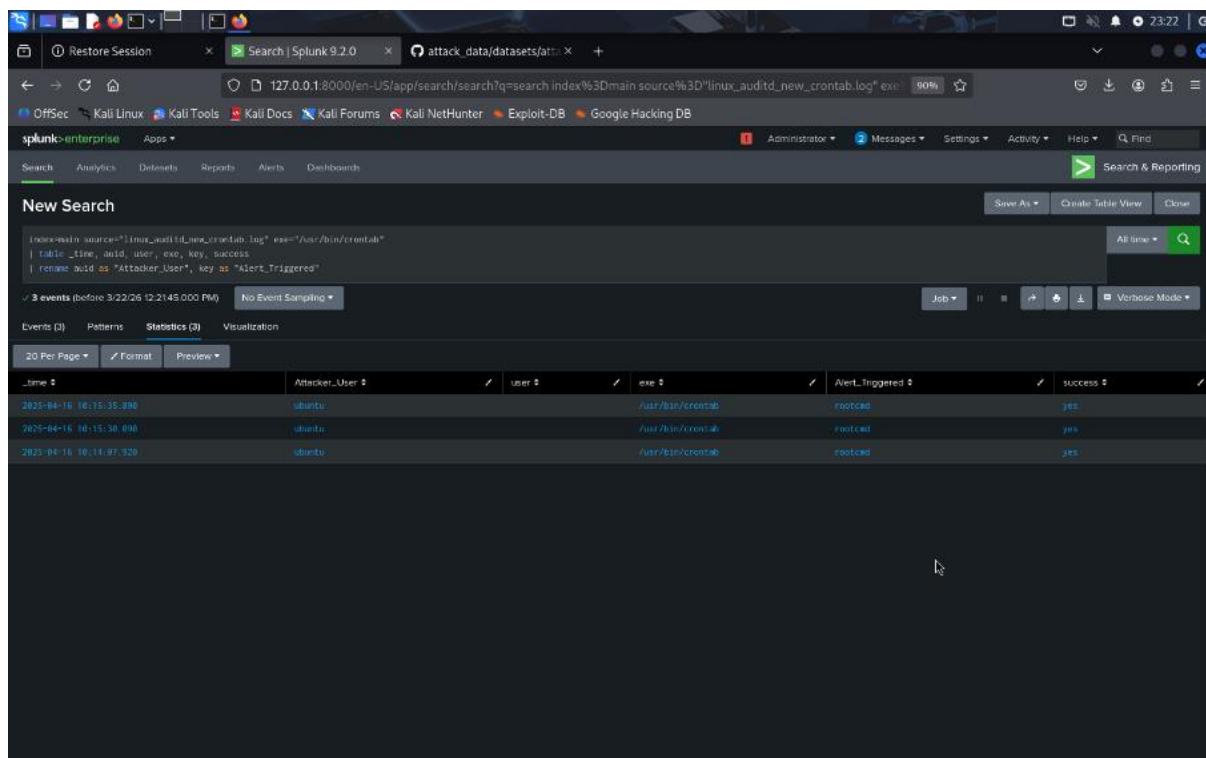


These activity are malicious because a low-privileged user (ubuntu/1000) is successfully modifying the root account's scheduled tasks. By creating an unauthorized crontab entry, the attacker achieves persistence, ensuring their malicious payload executes with full administrative privileges every time the system boots or at recurring intervals.

The query:

```
'index=main source="linux_auditd_new_crontab.log"
| table _time, audit, user, exe, key, success'
```

Screenshot B



The malicious activity triggered the '**rootcmd**' security alarm. By gaining unauthorized access to the root crontab, the attacker establishes **persistence**, ensuring their malicious scripts run with full administrative privileges every time the system reboots. The query has been used :

'index=main key="rootcmd" auid=1000 | table _time, auid, user, exe, key, success'

Conclusion

The Logic

In Linux systems, User IDs (UID/AUID) from 0 to 999 are reserved for system services and daemons. Any ID 1000 or higher is assigned to human users like ubuntu in our case. Under normal operations, system services rarely "interactively" edit crontabs. Therefore, any instance of a human user (auid >= 1000) modifying a crontab especially the root user's crontab is highly anomalous and should be treated as a potential persistence attempt.

Refined Detection Rule (SPL)

This query filters for the crontab executable and triggers an alert if the person running it is a standard user, specifically focusing on successful modifications.

```
index=main exe="/usr/bin/crontab" auid>=1000 success=yes  
| stats count by _time, auid, user, host, key  
| rename auid as "Detected_User", key as "Security_Rule_Trigger"
```

Security Impact

Implementing this rule allows a Security Operations Center (SOC) to catch attackers in the Installation/Persistence phase of the Cyber Kill Chain. By detecting this early, an organization can prevent an attacker from maintaining a "foothold" in the environment.